# DATA QUALITY AUDIT & ANOMALY REPORT

*Comprehensive Ingestion Profiling & Quality Control Framework | Task 01*

**MANDATORY DATA AUTHENTICITY NOTICE: SIMULATED DATA -- CREATED FOR ANALYTICAL TESTING**

VELOOP Rewards is an active developing product. All metrics, user records, and event logs analyzed herein are synthetic test cohorts created strictly to evaluate data readiness, schemas, and BI pipelines. They do not represent actual commercial users, revenue, or business performance.

## 1. Executive Quality Audit Overview

This audit profiles the raw dataset 'Users_1008_Messy' (1,008 rows x 28 columns) against strict enterprise data quality dimensions: Completeness, Uniqueness, Validity, Accuracy, Consistency, and Timeliness. The overall computed Data Quality Index (DQI) is 71.4%, requiring immediate data engineering remediation prior to production ingestion.

## 2. Identified Anomaly Taxonomy (Users_1008_Messy)

| Anomaly Classification | Rule Violated | Defect Count | Severity | Remediation Action |
|---|---|---|---|---|
| Malformed Email Syntax | RFC 5322 Standard | 54 rows (5.4%) | HIGH | Client-side regex gating & email verification |
| Negative XP Balances | CHECK (xp_balance >= 0) | 42 rows (4.2%) | CRITICAL | Database CHECK constraint & double-entry ledger |
| Future Timestamps | Timestamp <= NOW() | 38 rows (3.8%) | HIGH | NTP server time synchronization at ingestion |
| Missing Country Codes | ISO 3166-1 Alpha-2 | 48 rows (4.8%) | MEDIUM | IP-based GeoIP fallback enrichment pipeline |
| Colliding User IDs | PRIMARY KEY Uniqueness | 8 rows (0.8%) | CRITICAL | Migration to UUIDv4 / ULID auto-generation |
| Out-of-Range Ages | 18 <= Age <= 100 | 44 rows (4.4%) | MEDIUM | Enforce date-of-birth picker with age validation |

## 3. Quality Control Framework & Prevention Policies

* **Pre-Ingestion API Gating:** Enforce strict JSON schema validation at the ingestion gateway, immediately rejecting non-conforming payloads with HTTP 422 Unprocessable Entity.
* **Automated Dead-Letter Queue (DLQ):** Route malformed client payloads into an isolated SQS/Kafka Dead-Letter Queue for asynchronous inspection rather than polluting analytical tables.
* **Automated Regression Assertions:** Embed 5 core SQL quality assertions (06_data_quality_audit_checks.sql) into the CI/CD pipeline, halting builds upon detection of negative balances or orphan records.